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Do Credit Supply Shocks Affect Employment in Middle-Income Countries?†

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This paper studies the effect of bank credit supply shocks on formal employment in Mexico using a proprietary dataset containing information on all loans extended to firms by commercial banks during 2010–2015. We find large impacts on the formal employment of small and medium firms: a positive credit shock of 1 standard deviation increases yearly employment by 1.4 percentage points. The shares of uncollateralized credit and credit received by family firms, younger firms, and firms with no previous bank relationships also increase, suggesting that credit shocks may play a more prominent role for employment creation in credit-constrained settings. (*JEL* D22, G21, G32, J23, L25, M51, O16)

Small and medium-sized enterprises (SMEs) have attracted much interest from both scholars and policymakers since they play a critical role in the provision of employment around the globe (Ayyagari, Demirgüç-Kunt, and Maksimovic 2012). Yet, the ability of SMEs to create jobs may be hampered by their limited access to adequate financing, particularly in low- and middle-income economies (LMIEs), where credit constraints are considerably more severe than in advanced economies (AEs) (Beck, Demirgüç-Kunt, and Maksimovic 2004; Ayyagari, Demirgüç-Kunt, and Maksimovic 2012; Stein, Pinar-Ardic, and Hommes 2013). With lower saving rates, weaker investor protection, underdeveloped credit bureaus, and less competitive banking environments, the credit markets of LMIEs are smaller in size and deal less efficiently with informational asymmetries (La Porta et al. 1997; Djankov, McLiesh, and Shleifer 2007; Calomiris et al. 2017). Thus, SMEs in these countries face higher interest rates, larger credit constraints, and greater costs of switching across banks. This implies, however, that they may also be more likely to benefit from positive credit supply shocks, in turn creating more job opportunities for workers in the local market where they operate. Using a similar empirical strategy to Greenstone, Mas, and Nguyen

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(2020) for the United States, in this paper we study the impact of bank-specific credit shocks on SME employment in Mexico, finding relatively large effects. We further explore heterogeneity by firm and credit types that shed light on why credit shocks may have larger effects on employment in this context.

The relationship between credit growth and aggregate economic outcomes has been well documented in the macro literature, which shows that these variables tend to comove for both AEs (Apostoaie and Percic 2014; García-Escribano and Han 2015; Azariadis 2018) and LMIEs (García-Escribano and Han 2015). Identifying a causal link between credit and these outcomes is, however, a difficult empirical task, given obvious concerns about omitted variables and reverse causality. Early studies in the micro literature investigated the impact of government-induced bank branch expansions on economic activity, competition, and rural poverty, using difference-in-difference techniques that exploit variation across locations and over time (Jayaratne and Strahan 1996; Black and Strahan 2002; Burgess, Pande and Wong 2005).

More recently, an increasing availability of detailed data on credit (at the firm or loan level) has prompted a surge of papers that construct credibly exogenous measures of credit supply shocks to study their impact on employment (Greenstone, Mas, and Nguyen 2020; Chodorow-Reich 2014) and on other real outcomes such as investment and exports (Fraisse, Lé, and Thesmar 2019; Amiti and Weinstein 2018; Berton et al. 2018; Chodorow-Reich and Falato 2022).[1] However, most of these studies focus on the United States or other AEs for which these datasets are available. As observed above, credit supply shocks may have larger employment impacts in LMIEs, given that firms in these economies are more likely to be credit constrained and face worse credit conditions.

Our study contributes to this literature by measuring the causal relationship between credit supply shocks to SMEs (fewer than 250 employees) and changes in employment in the Mexican context. To this end, we exploit an extraordinarily detailed dataset containing loan-level information for the universe of loans extended by commercial banks to private firms in Mexico, together with publicly available data on the total number of employees reporting formal wages to the Mexican Social Security Institute (Instituto Mexicano del Seguro Social, IMSS). Our study period spans from 2010 through 2015, a time when the Mexican economy was relatively stable and credit availability to firms increased. To our knowledge, this is one of the first papers to study the effect of credit supply shocks on the formal employment of SMEs in a developing country, and one of the few to explore the relevance of credit supply shocks during normal times.[2]

[1] See Guler et al. (2021) for a review of the literature on the real effects of bank credit supply shocks using microdata.

[2] In a related paper, Morais et al. (2019) estimate, for a sample of relatively large Mexican firms, the employment effects of foreign monetary policy shocks affecting banks in Mexico linked to European and US banks. They find that firms more exposed to increases in foreign interest rates saw a small decline in employment levels, which they acknowledge to be a lower-bound estimate in that the effect is expected to be greater for small firms (which are not well represented in their sample). Their analysis complements ours in its focus on external monetary policy shocks on relatively large firms, while we analyze the effect of regional credit supply shocks on small and medium firms. A recent paper by Acosta and Cortés (2022) exploits the same dataset as that employed herein to estimate the relationship between credit and employment. The main difference between their study and ours is that they use the repayment of bank loans by local governments as the source of variation in credit supply at the local level. Aside from potential concerns regarding whether their instrument satisfies the exclusion restriction, our study is better positioned to explore differences in the impact of credit on employment between Mexico and the United States.

Specifically, we take advantage of differences in preexisting bank market shares across Mexican labor markets, along with variation over time in total lending from each bank, to empirically assess the causal link between credit supply shocks and changes in employment. We follow Greenstone, Mas, and Nguyen (2020) and construct our explanatory variable in two steps. First, we estimate the changes in lending from each bank that cannot be explained by variations in credit demand across labor markets. Second, we calculate labor market–level weighted averages of these bank-credit shocks. To build these averages, we use preexisting market shares of each bank in each labor market as weights.[3] The estimated credit shocks are interpreted as the expected percentage change in credit supply in a labor market, considering the past market share of each bank in that market and the national-level bank-specific credit supply change. In essence, our empirical strategy asks whether labor markets with banks that have higher idiosyncratic increases in firm credit availability also experience larger employment changes. A series of checks suggest that these shocks are likely exogenous to local market conditions.

We find that, for the Mexican context, credit supply shocks have a large and significant impact on formal employment in SMEs. Specifically, we find that in a labor market with a positive credit shock of 1 standard deviation, yearly employment increases 1.4 percentage points more than in other labor markets with no shocks (22 percent of the mean yearly employment change in our sample). Moreover, the richness of our data also allows us to study the type of credit and firms most affected by credit shocks, which contributes to understanding the economic forces behind the large effect on employment. We find that credit supply shocks are associated with increases in loans without collateral and in loans to new firms, family firms, and firms without recent credit history—that is, enterprises that are more strongly subject to informational asymmetries and credit constraints. We also show that our results are robust to various other specifications and robustness exercises.

Our outcomes contrast with the moderate to null employment effects observed in recent work that focuses on the relationship between credit and employment in AEs during normal times. Many papers use firm-level data from developed countries to study the relationship between employment and credit supply either during the 2009 financial crisis or in an abrupt negative financial shock, and find evidence of moderate negative effects on employment in such periods (Barbosa, Bilan, and Célérier 2019; Bentolila, Jansen, and Jiménez 2017; Berton et al. 2018; Boeri, Garibaldi, and Moen 2013; Caggese, Cuñat, and Metzger 2019; Chodorow-Reich 2014; Giannetti and Simonov 2013; Hochfellner et al. 2015; Huber 2018; Mehrotra and Sergeyev 2021; Moser et al. 2022; Popov and Rocholl 2018; Sharma and Winkler 2018; Siemer 2019). However, evidence on the effect of credit supply shocks on employment in

Specifically, our measures of credit supply shocks are constructed as in Greenstone, Mas, and Nguyen (2020), who find no effects on employment during normal times in the United States.

[3]The construction of shift-share measures of this kind and their defense as a source of arguably exogenous variation in a regressor of interest was first proposed by Bartik (1991). These have been exploited in a variety of settings, including, for example, by Aizer (2010), who explores the impact of gender wage gaps on domestic violence; Autor, Dorn, and Hanson (2013) and Dell, Feigenberg, and Teshima (2019), who measure the impact of Chinese competition on US labor market outcomes and Mexico's homicide rates, respectively; Card (1990), who estimates the elasticity of substitution between native and migrant labor; and Gutierrez and Teshima (2018), who measure the impact of import competition on firm-level environmental outcomes.

normal periods is more limited. One exception is Berg (2018), who uses a lender cutoff rule on credit scores to analyze the effect of loan rejections on the assets and employment of European firms during the years following the Great Recession. This study finds no effects on employment in general, but moderate negative effects of loan rejections on employment for small and illiquid firms that are unable to borrow from other banks and, therefore, hold more cash as precautionary savings.

Another exception is Greenstone, Mas, and Nguyen (2020), upon which we base our empirical methodology. In that paper, credit supply shocks in the United States have no effect on the employment of small firms during the years prior to the 2009 recession despite the fact that credit supply was growing rapidly. We explore different potential economic and methodological reasons for the discrepancy between our outcomes and those of Greenstone, Mas, and Nguyen (2020), including differences in the institutional framework of developing countries and AEs, and the role of labor informality. We argue that the magnitude of the effect is likely driven by the fact that SMEs face important credit constraints in Mexico (which may also hold in other LMIEs) and may thus benefit more from a bank-specific idiosyncratic credit supply shock than in the United States.

The remainder of the paper is organized as follows. Section I discusses the importance of better understanding the effect of credit shocks on employment in the Mexican context. Section II describes the data and the definition of the unit of observation. Section III outlines the empirical strategy. Section IV presents the results exploiting administrative data. Section V explores the reasons behind the discrepancy between our results and those of AEs. Section VI presents robustness checks, and Section VII concludes.

I. Institutional Background

In belonging to the group of LMIEs, Mexico offers an interesting case for studying the impact of credit supply shocks on the employment of SMEs. Credit markets are narrower and less efficient in LMIEs than in rich nations, and therefore usually provide less favorable financing conditions to firms. Positive credit supply shocks could accordingly have larger effects on easing the credit constraints of SMEs in LMIEs, and thus generate larger employment effects than in AEs.

The literature identifies several reasons for why credit markets are underdeveloped in LMIEs. One explanation relates to differences in GDP per capita. In having smaller GDP per capita, LMIEs have lower saving rates (Masson 1998; Loayza, Servén, and Schmidt-Hebbel 2000), which means they have fewer domestic resources available for intermediation and, consequently, smaller-sized credit markets. In Mexico, for instance, the ratio of private credit to GDP is only a quarter of the same ratio in the United States or, on average, the OECD countries (34.6 and around 144.6 percent, respectively).

Credit markets in LMIEs also deal less efficiently with informational asymmetries and channeling financial resources to firms. When lenders are uncertain as to a borrowers' ability to repay their loans, they frequently charge higher interest rates and reduce credit availability, generating credit constraints. This is especially relevant for SMEs (whose ability to repay is relatively more uncertain) and for loans

without collateral, for which the lender does not recover any of the loss in the case of a default. Similarly affected are young firms and those that do not have a credit history that can inform lenders on their recent performance and, therefore, on their probability of default (see Galindo and Schiantarelli 2002). Family firms may also suffer from credit constraints in that they tend to be smaller in size than nonfamily firms and may have less access to collateral, as discussed in detail in Section IIA.

While these informational asymmetries are common to all countries, the institutional frameworks in LMIEs are less prepared to deal with them. Lenders in these countries find it harder to force repayment or grab collateral because their legal investor protection is weaker; it can also be more difficult to obtain information on borrowers, as their credit registries are underdeveloped (La Porta et al. 1997; Djankov, McLiesh, and Shleifer 2007; Calomiris et al. 2017). These characteristics also explain, in part, why the ratio of private credit to GDP is lower in LMIEs (Djankov, McLiesh, and Shleifer 2007).

A less competitive banking environment represents an additional reason why credit conditions are worse for SMEs in LMIEs. While the link between market structure and credit conditions is unknown in principle, it has been shown that bank concentration increases credit constraints (Beck, Demirgüç-Kunt, and Maksimovic 2004). This effect is only found for small firms and for countries in which the institutional framework is sufficiently weak, such as LMIEs. Furthermore, and important for the purposes of this paper, a less competitive environment, along with lower saving rates and a weaker institutional framework, could result in higher costs of switching across banks for firms.

Mexico provides a great context for analyzing the impact of credit supply on employment in an LMIE. While lower than some Latin American countries', Mexico's ratio of private credit to GDP resembles that of several other LMIEs, including Albania (33.0 percent) and Indonesia (38.8 percent).[4] Further, 28.3 percent of SMEs in Mexico are credit constrained (Kuntchev et al. 2013), consistent with other LMIEs.[5] The Mexican National Institute of Statistics and Geography's (Instituto Nacional de Estadística, Geografía e Informática, INEGI) 2019 Economic Census similarly shows that 36.8 percent of Mexican firms were in need of external financing in 2018 but did not use banking credit, among other things, because their applications were rejected or because the offered interest rate was too high.[6] Moreover, credit rationing is a problem experienced mostly by SMEs: only 3 percent of large firms (more than 250 employees) report being credit constrained. Firms in Mexico also find it costly to switch banks: credit access depends on the closeness of the relationship between firms and banks (Galindo and Schiantarelli 2002) and the length of firm-bank relationships mitigates credit rationing to firms (De la Garza and Lopez-Videla 2015).

[4] For instance, Chile has a private credit to GDP ratio of 99 percent, Brazil's is 52.8 percent, and Colombia's is 43.8 percent.

[5] Kuntchev et al. (2013) define a firm to be credit constrained if (i) its applications were rejected or it did not apply for loans even though it needed external finance, or, (ii) despite having previously used external finance, the firm did not apply for loans even though they were in need of external financing, or applied for loans but were rejected.

[6] In the INEGI census, firms that reported not having used banking credit for any reason other than "not needing it" are considered to be firms that did not use banking credit even though they needed external financing.

In addition to these credit market features, the distribution of employment in Mexico further makes this an interesting setting for the question at hand. Unlike most AEs, SMEs account for a significantly larger portion of employment and employment growth. Specifically, INEGI's 2019 Economic Census shows that SMEs in Mexico, defined as firms with at most 250 employees, accounted for 71 percent of total employment in 2018. In contrast, in the United States, for example, data from the US Bureau of Labor Statistics indicate that firms in this same category accounted for only 45 percent of total private employment over the first quarter of 2019. Hence, an improvement in credit conditions for SMEs in Mexico has the potential to generate comparatively larger employment effects.

In summary, the institutional framework in Mexico makes it more difficult for firms and banks to overcome informational asymmetries than it would be in a developed economy, resulting in credit rationing and poorer credit conditions, particularly for SMEs. Moreover, the lower ability to deal with asymmetric information and the lower level of bank competition increases the costs for firms of switching banks, making them more vulnerable to bank-specific shocks.[7] The empirical strategy adopted herein is thus particularly well suited for addressing our research question, where we expect that a positive credit supply shock has the potential to generate larger employment impacts in Mexico than in more developed economies, such as the United States.

II. Data

A. *Data Sources*

Our analysis combines four sources: a dataset including detailed information on the universe of new credit lines issued by private banks to firms in Mexico maintained by the central bank (Banco de Mexico, Banxico), publicly available data on formal employment published by the Mexican Social Security Institute (IMSS); the Mexican Employment Survey (ENOE), and information from the 2010 census. A detailed description of all the datasets (main and secondary data) is presented in online Appendix Table A1.

Credit Data.—All commercial banks are mandated to submit monthly reports to the regulator, the National Banking and Values Commission (Comisión Nacional Bancaria y de Valores, CNBV), with detailed information on all new and existing loans extended to firms regardless of firm and loan size. A wide variety of characteristics are reported for each loan. Importantly for the purposes of this study, for each loan we observe the number of employees and location (municipality) of the receiving firm, the issuing bank, and the loan size.

From these reports that are housed at the central bank (Bank of Mexico 2020a), we build two datasets. The first contains information at the labor market-bank level, which we use to construct our measure of supply credit shocks. The second

[7] Note that our empirical strategy, using Bartik-type measures of credit supply shocks, is based on bank-specific shocks aggregated at the labor market level.

contains information at the labor market level that we use to explore the relationship between loan originations and supply shocks. For both datasets, we restrict the analysis to loan originations granted to private firms with less than 250 employees and aggregate the data at the yearly level.[8] The yearly aggregation responds to the fact that for some labor markets, relatively few credit lines are issued in each year, meaning that monthly data contain many observations (labor markets) with zero credit. Aggregating at the yearly level allows us to obtain nonzero measures of credit during the entire period analyzed for a total of 468 labor markets, corresponding to 97 percent of the Mexican labor force.[9]

From the data, we observe 47 different banks that report having issued at least one credit during 2010–2015. When constructing the first dataset, we impose two sample restrictions to ensure comparability. First, as is standard in the literature, for each group of banks that participated in a merger, we recode the banks' identifiers to correspond to a single bank throughout the entire period (leaving us with 44 banks in total). Second, we restrict the sample to banks that report having issued at least one credit line to firms in at least two labor markets each year in the data (corresponding to 23 of those 44 banks). All remaining loans are assigned to a single "other banks" identifier. The second dataset simply contains the sum of all new loan amounts granted to SMEs each year by all banks in the reports.

The data also allow us to distinguish between new credit with and without collateral and whether the firm is family owned or not (i.e., corporations). Family firms—which represent 37 percent of the credit in our sample—are smaller in size, have less collateralized credit, face a higher interest rate, have higher nonperforming loan rates (default rate), and have access to a lower share of open credit lines known as revolving credit (see online Appendix Table A2). For nonfamily firms, we classify them as either young (less than five years since they were established) or old (more than five years).[10] Finally, because we observe the universe of firms with loans, we can also differentiate between firms with and without credit during the previous year as a proxy of attachment to the financial system and recent credit history.

Formal Employment.—To analyze the impact of credit on formal employment, we exploit a publicly available dataset published monthly by the IMSS (2020) that contains municipality-level information on the total number of registered workers

[8] We exclude loans granted to the local and federal governments, as well as those to firms operated by the public sector. Firms report their number of employees in their loan application, and this is updated each time they are granted a new credit line. To reduce measurement error, we consider firm size as the median number of employees reported by each firm throughout the entire period for which we have data.

[9] As we explain in greater detail below, in our empirical strategy, the first step consists of measuring credit supply shocks using as input the national changes in new credit granted by each bank. We estimate this in an equation that decomposes labor market–level changes in credit into labor market and bank-specific components (equation (1)). To precisely estimate the bank-specific component, it is important to have enough data on new credit granted by each bank in different local labor markets during the reference period. The shorter the considered period, the more observations with no observed change in new credit (bank–local labor market cells with zero values are dropped from the estimation), and the noisier these banks specific components become, to the point that the credit supply shocks estimated with monthly and even quarterly data are no longer informative (e.g., the shocks are not related with observed changes in new credit).

[10] This information is only available for nonfamily firms, for which the firm identifier contains the year in which the firm was created. For family firms, the firm identifier does not include any information relative to the year it first began to operate.

by firm size, sector, and worker characteristics (such as gender and wage categories). The IMSS is the institution responsible for providing health-care services to formal workers in the private sector. Since all private firms in Mexico are required to contribute a fraction of each of their employees' formal salary to the IMSS, this institution has access to detailed information on all workers in private firms.

From these data, we extract the total number of employees in the low- and high-wage brackets (less or more than two minimum wages) as well as average salaries in SMEs. We rely on the data for December of each year and aggregate municipalities up to the labor market level. SMEs (i.e., firms with less than 250 employees) cover around 80 percent of the universe of formal workers. Measuring total employment at the end of the year allows us to estimate the impact of the average credit shock during the year on total employment at the end of the period, allowing for the possibility that the effects of credit shocks on employment are not immediate.[11] We also complement this approach by exploring potential dynamic effects in greater detail through auxiliary biannual analyses that include the contemporaneous credit shocks and two of their lags.

Household Survey: ENOE.—To discern whether our results on formal employment can be explained by workers shifting from the informal to the formal sector, we exploit information from the publicly available Mexican Employment Survey (ENOE) from INEGI (2020a). For comparability, we employ ENOE information for the fourth quarter of each year over the 2010–2015 period, and calculate the number of formal and informal employees (and the sum of both) in SMEs in each of the labor markets covered by the survey. To maintain consistency with the definition of formality adopted when exploiting the IMSS data, we define formal workers as those who declare being affiliated with the IMSS. Informal workers are thus those who do not report an affiliation to any of the formal social security subsystems in Mexico (IMSS, ISSSTE, and PEMEX). Unfortunately, the results obtained with the ENOE information are not fully comparable to those obtained with the IMSS administrative data, as the ENOE sample primarily covers large, urban localities.[12] Therefore, in presenting these results, we also replicate our main analysis with IMSS data but restrict our attention to labor markets containing the ENOE urban localities.

Census 2010 Data.—Finally, we construct a set of labor market characteristics that we use to elucidate the validity of our identification strategy. Specifically, we employ the 2010 census conducted by INEGI (2020b). From this census, we construct a series of socioeconomic indicators for each municipality (which we then aggregate to the labor market level), including the total population; the fraction of the population in different age groups (less than 15, 15 to 64, and 65 and over);

[11] Our results are similar if we use June instead of December as our target month to measure employment outcomes and are available upon request.

[12] Though highly correlated, survey and administrative data are not the same. Administrative data contain information on firms' location, while survey data contain information on workers' addresses. Because the survey data are limited to certain localities in urban centers and not to the total local labor market, these two datasets differ even after restricting the data only to those localities where the survey data are representative (workers from firms in the administrative data could live in different localities). Results from the survey data are expected to be noisier than the administrative data considering that firms' locations better identify the units affected by the credit supply shock.

the fraction of immigrants (population born out of state); the fractions of Catholic, indigenous, illiterate, and with completed primary and secondary education; the fraction of the population with health insurance; the fraction of households with a female head; and the fraction of households with access to running water, sewerage, and electricity.

B. *Definition of the Units of Observation*

To explore the impact of credit shocks on real outcomes, it is essential to identify the relevant units of analysis. Credit shocks to a particular firm (or region) may impact the real outcomes of other firms that are not directly affected by the shocks but, nonetheless, are related to the enterprise hit by the shock. In particular, firms that participate in the same labor market or have economic relationships (clients, suppliers, or competitors) with the affected firm are also likely to end up being affected by the shock (Herreño 2021; Barrot and Nanda 2020). To illustrate the intuition behind this argument, we begin this section by presenting a very simple example and then describe how we chose the units of analysis for our empirical estimation of the impact of credit shocks on formal employment.

Consider the case of two representative firms (A and B) that participate in the same competitive labor market but borrow from different banks. Assume further that firm A faces a borrowing limit that does not allow it to scale production and labor to the point where the marginal product of labor equals the market wage. Imagine now that a positive credit supply shock relaxes this borrowing limit (the bank with whom Firm A has a relationship becomes more able or more willing to lend to SMEs—e.g., due to a decline in nonperforming loans in portfolios such as consumer credit or mortgages), allowing this firm to expand its production and hire more labor. In equilibrium, the increase in labor demand pushes the wage up and could change firm B's choice of capital and labor, potentially inducing this firm to hire fewer workers.

In this example, a difference-in-difference estimation that compares the changes in employment of enterprises like firm A (who received the positive credit shock) to those of enterprises like firm B delivers a biased estimate of the credit shock on total employment. The direction and magnitude of the bias depends on the degree to which the firms are interrelated and participate in the same markets—i.e., in the example, there is an upward bias because firm B hires less workers. In other words, the stable unit treatment value assumption (SUTVA) is violated for firm B because its response depends on a credit shock (treatment) assigned to another firm that participates in the same local labor market.[13]

To adequately measure the impact of credit shocks on total employment, and to avoid bias, we must define the outcome variables (credit and employment) at the labor market level. This aggregation level enables us to pick up not only the effect for firm A but also that of all neighboring firms participating in the same labor market. The intuition is that local labor markets are the unit of analysis that best characterizes the

[13] A firm C could be positively affected via trade (supplier or purchaser of firm A), also violating the SUTVA condition but switching the direction of the bias.

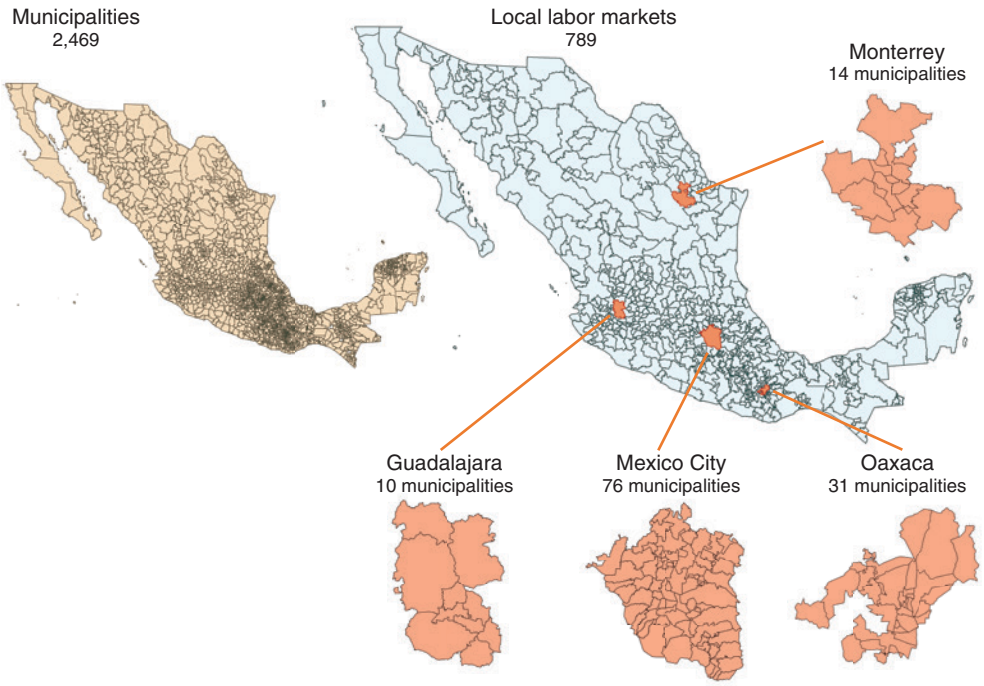

Figure 1. Local Labor Markets and Municipalities

Notes: Authors' classification of local labor markets following Blyde, Busso, and Romero (2020). Municipalities are assigned to different markets according to commuting data. The figure also shows in detail the labor markets of some of the largest urban centers in Mexico.

credit conditions faced by small firms, as well as their trade connections (location of suppliers, purchasers, and competitors).

We group municipalities into labor markets following the procedure implemented by Blyde, Busso, and Romero (2020). Their algorithm constructs 789 local labor markets based on commuting zones by grouping the 2,469 municipalities in Mexico (local administrative boundaries), and operates as follows: (i) it identifies 599 urban municipalities to which people from other municipalities commute daily for employment purposes and classifies them as "central," meaning that each of these municipalities is considered to be a separate labor market; (ii) it follows an iterative procedure whereby the remaining municipalities are added to each of these markets based on geographic distance and on whether their residents commute to a "central" municipality for work purposes; and (iii) the 190 municipalities that were not assigned to any central municipality after the iterative process are defined as separate, individual labor markets. The final grouping of municipalities into labor markets is illustrated in Figure 1.

C. *Sample Restrictions and Descriptive Statistics*

To have a balanced panel, we restrict our sample to the 468 labor markets with nonzero observations for new credit lines and for the number of employees in SMEs. In all markets, more than one bank issued a loan during the period analyzed. Figure 2

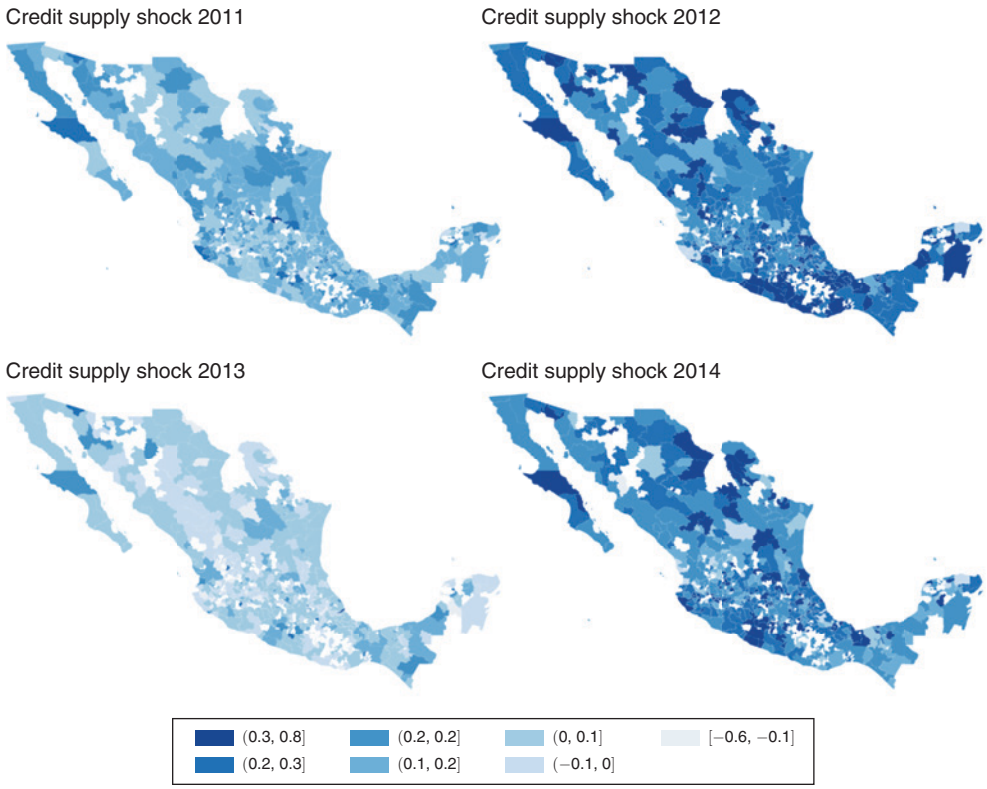

FIGURE 2. EXPOSURE TO SHOCKS FOR LOCAL LABOR MARKETS IN SAMPLE

Notes: Authors' calculations of credit shocks at the local labor market level according to equation (3). The figure shows the estimated credit supply shock by baseline year. The figure also shows the geographic location of labor markets in our restricted sample (labor markets excluded from the empirical analysis are shown in white). In all labor markets, more than one bank issued a loan during the period analyzed. Credit supply shocks are estimated using proprietary credit data.

displays the geographic location of the labor markets in our restricted sample (the figure shows the magnitude of the estimated shocks for different years in our sample), while Table 1 shows observable characteristics for these locations and for those excluded from the empirical analysis. Not surprisingly, the markets included in the analysis have a higher population, a larger fraction of the population aged 15 to 64, higher education levels, and relatively greater access to public services such as running water, sewerage, and electricity. Our final sample includes 60 percent of all labor markets as defined by Blyde, Busso, and Romero (2020), but concentrates 97 percent of the country's population.

Panel A in Table 2 presents descriptive statistics of the employment data at the labor market level. Consistent with national-level data, on average, in a labor market, 57 percent of employees in SMEs earn less than 2 minimum wages. Panel B presents descriptive statistics for the data on new credit to SMEs in labor markets included in our main analysis: the amount of new credit (millions of real Mexican pesos); the share of credit without collateral; the share of credit to family firms, to new firms, and to firms without credit the previous year; the number of banks in a

Table 1—Descriptive Characteristics of Local Labor Markets in the Sample

Variable	Not in sample	In sample	Difference
Population (thousands)	19.7	467.5	447.7
	(27.2)	(2,014.1)	(112.5)
Fraction of population under 15 years old	0.301	0.301	−0.000
	(0.055)	(0.031)	(0.003)
Fraction of population between 15 and 64 years old	0.584	0.614	0.029
	(0.045)	(0.027)	(0.003)
Fraction of population over 65 years old	0.106	0.075	−0.031
	(0.051)	(0.023)	(0.003)
Fraction of population born in state	0.922	0.866	−0.056
	(0.082)	(0.099)	(0.007)
Fraction of pop. who lived in a different state in 2005	0.022	0.025	0.004
	(0.023)	(0.022)	(0.002)
Fraction of pop. that speaks an indigenous language	0.261	0.075	−0.186
	(0.331)	(0.158)	(0.018)
Fraction of pop. that is illiterate	0.146	0.094	−0.052
	(0.092)	(0.056)	(0.005)
Fraction of pop. with completed elementary school (and no more)	0.214	0.189	−0.025
	(0.056)	(0.039)	(0.003)
Fraction of pop. with completed secondary school (and no more)	0.200	0.216	0.017
	(0.055)	(0.043)	(0.003)
Fraction of pop. with health insurance	0.367	0.332	−0.035
	(0.191)	(0.118)	(0.011)
Fraction of pop. with seguro popular	0.413	0.354	−0.060
	(0.204)	(0.153)	(0.013)
Fraction of pop. that is Catholic	0.858	0.854	−0.004
	(0.125)	(0.108)	(0.008)
Fraction of households with female head	0.223	0.229	0.006
	(0.060)	(0.036)	(0.003)
Fraction of households with dirt floor	0.156	0.076	−0.080
	(0.144)	(0.068)	(0.008)
Fraction of households with electricity	0.941	0.968	0.027
	(0.080)	(0.034)	(0.004)
Fraction of households with water	0.802	0.855	0.053
	(0.190)	(0.130)	(0.011)
Fraction of households with sewerage	0.664	0.854	0.189
	(0.265)	(0.131)	(0.014)
Observations	321	468	789

Notes: Authors' estimation based on the 2010 census information aggregated at the local labor markets identified by Blyde, Busso, and Romero (2020). The table shows the descriptive statistics for local labor markets in and out of our main sample as well as their statistical difference.

local labor market; and the estimated credit supply shocks. Importantly, a large fraction of the total new credit is not collateralized, and family firms receive 37 percent of credit within a labor market.

Panel C presents statistics aggregating the credit data at the bank level: the number of new loans granted per bank in a year; the average amount lent (millions of real Mexican pesos); the share of new credit to firms without a bank relationship and to firms with another bank relationship; the share of firms related to only one bank in a given year; and the number of local labor markets to which the bank has granted a loan. Consistent with firms in Mexico having limited opportunities to "shop around"

Table 2—Summary Statistics

Descriptive characteristics	Mean	SD
Panel A. Formal employment characteristics		
Employment	18,727	100,755
Low-wage employment	8,914	42,187
High-wage employment	9,813	58,847
Share of low-wage employment	0.57	0.13
Panel B. Credit characteristics by local labor markets		
New credit (millions)	850	6,432
Share of credit without collateral	0.79	0.26
Share of credit to family firms	0.37	0.32
Share of credit to new firms	0.17	0.25
Share of credit to firms without credit the previous year	0.23	0.24
Number of banks in a local labor market	5.86	4.03
Credit supply shocks	0.11	0.16
Panel C. Bank-level credit data		
New loans per bank in a year	8,994	13,915
Average amount lent (millions real Mexican pesos)	4.94	4.27
Share of firm credit to firms without a bank relationship	0.28	0.34
Share of firm credit to firms with other bank relationship	0.06	0.09
Share of firm credit to firms related to only one bank in given year	0.74	0.15
Number of local labor markets per bank	159	146

Notes: The table shows the descriptive statistics for the sample used in the analysis for the 468 local labor markets (panel A and panel B), as well as descriptive statistics related to the credit data constructed at the bank-year level (panel C). Data on formal employment in Panel A come from IMSS data, and data from credit (panels B and C) come from proprietary credit data. Credit to firms without a bank relationship refers to firms that did not have any bank credit during the previous year. Credit to firms with other bank relationship refers to new credit to firms that had a relationship with another bank during the previous year. Credit to firms related to only one bank during the previous year indicates credit to firms that only obtain new credit with one bank in a given year.

for credit from different banks, less than 6 percent of credit is granted to firms that had a relationship with a different bank in the previous year, and 74 percent of firms have a banking relationship with only 1 commercial bank.

III. Empirical Strategy

A. *Credit Shocks*

Following Greenstone, Mas, and Nguyen (2020), we use a modified Bartik-type instrument to construct credit shocks that are arguably exogenous to local credit conditions. Specifically, we start by estimating an equation that decomposes labor market–level changes in credit into labor market– and bank-specific components. To this end, we run a set of separate regressions (one for each year $t \in \{1,2, \ldots, T\}$ in our data) of the following form:

$$\Delta \ln(Q_{m,b,t}) = \alpha_{m,t} + \gamma_{b,t} + \epsilon_{m,b,t}, \tag{1}$$

where the outcome variable is the log change in all new credit originations from bank b in labor market m between each pair of consecutive years t and $t-1$, $\alpha_{m,t}$ denotes labor market fixed effects in year t, $\gamma_{b,t}$ denotes bank fixed effects in year t, and $\epsilon_{m,b,t}$

is an error term. The labor market fixed effects capture the average change in lending to a labor market by all banks in our data. Hence, the bank-time fixed effect $\gamma_{b,t}$ is an estimate of the change in the credit supply of bank b that is not explained by changes in the economic conditions of labor market m that affect lending from all banks in the same magnitude (in percentage points). In this sense, $\gamma_{b,t}$ can be interpreted as the idiosyncratic credit shock of bank b for period t.[14]

Defining $L_{i,b,f,m,t}$ as the amount of the new loan i granted by bank b to a firm f that is located in labor market m over period t, we first calculate the market share of each bank in each labor market in $t-1$:

$$s_{b,m,t-1} = \frac{\sum_f \left(L_{i,b,f,m,t-1}\right)}{\sum_b \sum_f \left(L_{i,b,f,m,t-1}\right)}. \tag{2}$$

Then, we use $s_{b,m,t-1}$ to calculate the local labor market–level predicted lending shock, $Z_{m,t}$, as a weighted average of the bank-specific shocks for each labor market:

$$Z_{m,t} = \sum_b \left(s_{b,m,t-1} \times \hat{\gamma}_{b,t}\right), \tag{3}$$

where $\hat{\gamma}_{b,t}$ are the bank fixed effects estimated for each period. Intuitively, to build $Z_{m,t}$ shocks, we aggregate banks' national percentage change in credit supply (purged of banks' differential exposure to regional-level variation in credit demand) using as weights the banks' regional shares before the shock. The shocks are therefore interpreted as the expected percentage change in credit supply during year t in each labor market m generated by the heterogeneous regional exposure to banks' national supply shocks.

B. *On the Validity of the Instrument*

A recent strand of the literature questions whether Bartik-type instruments can satisfy the exogeneity condition for the estimation of causal effects. Goldsmith-Pinkham, Sorkin, and Swift (2020) illustrate how, under some circumstances, the exogeneity assumption is analogous to interpreting, in our context, that the bank shares are exogenous. However, Borusyak, Hull, and Jaravel (2022) show that with multiple time periods, the exogeneity of the shocks can provide identification, even when shares are endogenous.

Taken together, these insights imply that we can interpret our estimates as causal if these conditions are satisfied, which, in our case, means that the credit supply shocks are uncorrelated with other observed or unobserved labor market characteristics. In particular, the shocks must be uncorrelated with labor market characteristics that can predict changes in the outcome variable. To partially test for the validity of this assumption, in what follows we explore whether our credit shocks are correlated with observable variables other than the banks' market shares.

[14] Note that these credit supply shocks are not necessarily centered on zero. In fact, in our sample, the average credit supply shock is 11 percent, indicating that bank credit supply was increasing during our period of analysis. Normalizing the estimated shocks (mean and standard deviation equal to zero and one, respectively) does not alter the results.

Table 3 compares observable characteristics for labor markets for which the average shock is above and below our calculated median average credit shock. Differences in means between the two groups are small. Of the 18 observable characteristics in our data, only 3 are significantly different from zero at a high (5 percent) confidence level. A linear regression at the labor market level of the median shock against all observable characteristics listed in Table 3 cannot reject that all the estimated coefficients are jointly zero. The three regional characteristics that are statistically significantly different from zero are the fraction of the population that is illiterate, the fraction of households with a dirt floor, and the fraction of households with water, and all our main regressions include linear trends interacted with these variables. Additionally, in Section VI we show that our results are robust to including linear trends interacted with all the regional characteristics included in Table 3. This exercise shows that credit supply shocks do not follow any particular trends across local labor markets associated with other underlying observable characteristics that could bias our estimates.

As outlined above, our identification strategy exploits the fact that banks varied their lending to SMEs year by year, and that bank market shares differ substantially across local labor markets. We can therefore exploit heterogeneity in the exposure of local labor markets to different banks using the shift-share instrument introduced above. The implicit assumption is that firms find it difficult to switch banks in response to a shock affecting their current lender (e.g., Table 2 shows that only 7 percent of new credit is directed to firms with a previous relationship with another bank). If this is the case, a positive credit supply shock to a subset of banks in a given region will increase the aggregate credit supply in that area, easing the credit constraints faced by the firms. Note also that the changes in bank credit supply are estimated after extracting the changes in a region that are common to all banks. Hence, the credit supply changes at the bank level are net of any change in regional credit demand, generating credit supply changes at the bank level that are arguably "exogenous" for the local labor market.

This still leaves the question as to what exactly is generating the heterogeneous behavior at the bank-year level that we exploit. To answer this, and to validate our research design, Table 4 correlates the bank supply changes to different bank-level characteristics, based on the idea that the bank's general performance during that year should correlate with the estimated change in credit supply. For example, if a bank benefits from a decline in the NPL ratio (nonperforming loans to total outstanding loans) in other credit segments, such as mortgages or consumption, the ensuing improvement in its balance sheet may increase the loans available to SMEs. We formally test these associations in Table 4, where we present the results of a bank-level regression correlating the estimation of banks' specific credit supply changes with several financial indicators that are commonly used in the finance literature.[15] Columns 1 to 8 show univariate regressions with bank and time fixed effects, and column 9 incorporates all indicators simultaneously. Our results show

[15] Data come from banks' balance sheet indicators published by the financial regulator (Comisión Nacional Bancaria y de Valores 2020).

Table 3—Baseline Labor Market Characteristics and Magnitude of Credit Supply Shocks

Observable characteristics by average shock size			
Variable	Below median shock	Above median shock	Difference
Population (thousands)	570.1	364.9	−205.2
	(2,751.7)	(733.5)	(186.2)
Fraction of population under 15 years old	0.300	0.302	0.002
	(0.030)	(0.032)	(0.003)
Fraction of population between 15 and 64 years old	0.614	0.613	−0.001
	(0.027)	(0.028)	(0.003)
Fraction of population over 65 years old	0.075	0.075	−0.000
	(0.023)	(0.024)	(0.002)
Fraction of population born in state	0.865	0.868	0.003
	(0.103)	(0.095)	(0.009)
Fraction of pop. who lived in a different state in 2005	0.026	0.025	−0.001
	(0.026)	(0.018)	(0.002)
Fraction of pop. that speaks an indigenous language	0.079	0.071	−0.008
	(0.170)	(0.145)	(0.015)
Fraction of pop. that is illiterate	0.088	0.100	0.012
	(0.054)	(0.057)	(0.005)
Fraction of pop. with completed elementary school (and no more)	0.188	0.190	0.002
	(0.040)	(0.038)	(0.004)
Fraction of pop. with completed secondary school (and no more)	0.218	0.214	−0.004
	(0.040)	(0.045)	(0.004)
Fraction of pop. with health insurance	0.326	0.338	0.012
	(0.110)	(0.125)	(0.011)
Fraction of pop. with seguro popular	0.348	0.360	0.012
	(0.153)	(0.153)	(0.014)
Fraction of pop. that is Catholic	0.862	0.847	−0.015
	(0.105)	(0.110)	(0.010)
Fraction of households with female head	0.227	0.232	0.005
	(0.035)	(0.037)	(0.003)
Fraction of households with dirt floor	0.068	0.083	0.015
	(0.065)	(0.069)	(0.006)
Fraction of households with electricity	0.969	0.968	−0.001
	(0.041)	(0.024)	(0.003)
Fraction of households with water	0.870	0.840	−0.030
	(0.122)	(0.137)	(0.012)
Fraction of households with sewerage	0.857	0.850	−0.007
	(0.137)	(0.125)	(0.012)
Observations	234	234	468

Notes: Authors' estimation based on the 2010 census information aggregated at the local labor markets identified by Blyde, Busso, and Romero (2020), and our estimates of the credit supply shocks. The table shows the descriptive statistics for local labor markets with average credit supply shock above and below the median credit supply shock in our sample, as well as their statistical difference.

that an increase in bank credit supply to firms is associated with several bank level characteristics.

First, there is a negative association with the NPL ratio of nonfirm credit (household credit and mortgages), consistent with a higher repayment rate from other credit, liberating more resources for firm credit. Second, we find a positive association with the passive interest rate—that is, the rate the bank offers to depositors. Banks could

Table 4—Bank Credit Supply Changes and Banks' Characteristics

	Dependent variable: banks' yearly changes in credit supply								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NPL ratio (except firm credit)	−8.741								−12.54
	(3.166)								(4.185)
NPL ratio firm credit		−2.200							1.910
		(1.863)							(2.719)
Passive interest rate			4.554						3.114
			(0.531)						(0.867)
Ratio reserves to capital				0.134					0.502
				(0.344)					(0.290)
log deposits					0.226				0.115
					(0.145)				(0.125)
Return to equity						−0.0329			−0.504
						(0.682)			(0.569)
Provision to nonperfoming loans (Coverage ratio)							0.003		0.002
							(0.001)		(0.0008)
Capitalization index								−0.006	−0.019
								(0.005)	(0.016)
Observations	143	143	143	143	143	143	136	143	136

Notes: Authors' estimation of correlations between bank's credit supply changes and bank characteristics. Banks' yearly changes in credit supply are estimated by equation (1). Data come from banks' public balance sheet information reported by the financial regulator (Comisión Nacional Bancaria y de Valores 2020). Regressions include bank fixed effects and year fixed effects. Columns 1 to 8 present results for univariate regressions introducing one variable at a time, and column 9 present results controlling for all variables simultaneously. The NPL ratio (except firm credit) refers to the ratio of nonperforming loans to total outstanding loans (performing and nonperforming) for all private credit except for firm credit (consumer credit and mortgages). The table shows the correlation between the credit supply changes at the bank level and bank characteristics. Observations are weighted by total outstanding loans to account for bank size. Standard errors are clustered at the bank level.

increase this rate to collect more deposits and, therefore, increase their credit supply. Finally, banks with more solid financial indicators have a higher credit supply. This can be seen in the positive association between bank shocks and the coverage ratio and the ratio of reserves to capital. An increase in these two indicators suggests that the bank is solidly positioned financially and is thus in a better position to increase the credit available to firms. Importantly, we find no association between the NPL ratio of firm credit and the changes in credit supply to firms. This is expected given that we are "purging" the changes in banks' firm credit supply of changes in local credit demand, which could also be related to firm performance in that region and the NPL ratio of firm credit.

As a final piece of evidence, Figure 2 shows the size of the estimated credit shock in each labor market for four different years. Looking across the panels, we see that the estimated shocks show large variation both across regions for a given year and over time within a labor market. This provides suggestive evidence that the estimated credit supply shocks are not consistently larger in specific regions, providing further support for the hypothesis that our credit shocks are not related to idiosyncratic regional characteristics.

C. *Estimating Equation*

Having constructed the supply-driven credit shocks at the local labor market level, we estimate regressions of the following form to recover their causal impact on employment outcomes:

$$\ln\left(Outcome_{mt}\right) = \beta_1 Z_{m,t} + \delta_m + \pi_t + \varepsilon_{m,t}, \tag{4}$$

where $\ln\left(Outcome_{mt}\right)$ is the outcome variable in labor market m for period t (total new credit and formal employment in SMEs), the $Z_{m,t}$ variables are credit supply shocks at the local labor market level as defined above, δ_m are labor market fixed effects, π_t are period fixed effects, and $\varepsilon_{m,t}$ is an error term. This specification allows us to identify the effects of the credit supply shocks that took place during a given year on the end-of-the-year employment outcomes. The coefficient of interest is β, which can be interpreted as the yearly percentage change in formal employment in SMEs associated with a 1 percent change in new credit to small firms. Additionally, all our main regressions include linear trends interacted with the three municipality characteristics that are statistically significantly different from zero in Table 3—namely, the fraction of the population that is illiterate, the fraction of households with a dirt floor, and the fraction of households with water. In Section VI, we show that our main results are robust to not including these controls and to controlling for linear trends interacted with all the municipality characteristics in Table 3.

IV. Results

A. *Credit Shocks and Credit Originations*

For our empirical strategy to correctly identify the causal relationship between credit and formal employment, the credit shocks must not only be exogenous to local labor market level conditions (which we partially test in Section IIIB) but also predict credit at the local level. We provide evidence of this relationship in Table 5, which presents the results from regressing the observed log of new credit granted in each labor market in a given year against our shock measures. Column 1 includes only the contemporaneous shock as a regressor. To explore whether there are dynamic effects of credit shocks on new credit, columns 2 and 3 additionally include the shock in the previous year and in the previous two years as controls, respectively. The sample changes across specifications because for each lag added, we lose one year of data (online Appendix Table A3 shows results for the same sample across all specifications using the period 2013–2015). We also graphically present the estimated relationship between our measure of credit supply shocks and the log of new credit in Figure 3, panel A.

The estimated coefficients for the contemporaneous shock across specifications are large, positive, and significantly different from zero at a high confidence level. Our measure of supply-driven credit shocks are good predictors of new credit to SMEs in local labor markets. The estimate is also stable across specifications, indicating that shocks are relatively independent over time and that the coefficient associated with the

Table 5—Effect of Credit Supply Shocks on Loan Originations at the Local Labor Market Level

Dependent variable: log of total new credit to small firms			
	(1)	(2)	(3)
Credit shock$_t$	0.413	0.497	0.639
	(0.187)	(0.212)	(0.251)
Credit shock$_{t-1}$		0.212	0.194
		(0.168)	(0.232)
Credit shock$_{t-2}$			−0.0810
			(0.207)
Observations	2,340	1,872	1,404

Notes: Authors' estimation of equation (4) for the log of loan originations to small firms in local labor markets as the dependent variable. Data come from new credit and credit shocks come from proprietary credit data. Regressions include local labor markets and year fixed effects. The regression also includes linear trends interacted with the local labor market controls (fraction of illiterate population, fraction of households with dirt floor, and fraction of households with water). The table shows point estimates of the effect of credit supply shocks on loan originations. Observations are weighted by the number of workers in small firms in each labor market–year cell in 2010. Standard errors are clustered at the local labor market level.

contemporaneous shock in column 1 is not picking up any dynamic effect. In terms of economic significance, the average credit shock in our sample is 0.11, meaning that shocks during this period were positive on average, and its standard deviation is 0.16. Thus, in a local labor market receiving a positive shock of 1 standard deviation, new credit to SMEs increases by 6.61 percent ($0.16 \times 0.413 \times 100$) with respect to labor markets with no credit shock. The results are consistent with the hypothesis that it is costly for firms in Mexico to switch banks. Thus, bank-firm relationships tend to be stable, and firms respond quickly and strongly to temporal changes in the credit conditions offered by their traditional bank. It is also consistent with credit supply shocks facilitating access to the financial system for firms that do not have a previous bank relationship, such as new firms and those that do not have a recent credit history.

B. *Credit Shocks and Formal Employment*

Table 6 shows the reduced-form results using formal employment in SMEs as the dependent variable. As in the previous table, the sample changes across specifications because for each lag added, we lose one year of data (online Appendix Table A3 shows results for the same sample across all specifications using the period 2013–2015). Column 1 includes only the contemporaneous shock as a regressor. Columns 2 and 3 additionally include the shock in the previous year and in the previous two years as controls, respectively.

The coefficients associated with the contemporaneous shock are positive and significantly different from zero at a high confidence level, as well as stable across specifications. Unlike Greenstone, Mas, and Nguyen (2020), we find strong effects of credit shocks on formal employment for small firms in the Mexican context during normal times. The point estimate in column 1 indicates that a positive credit shock of 1 standard deviation (0.16) generates a 1.42 percentage point larger increase in

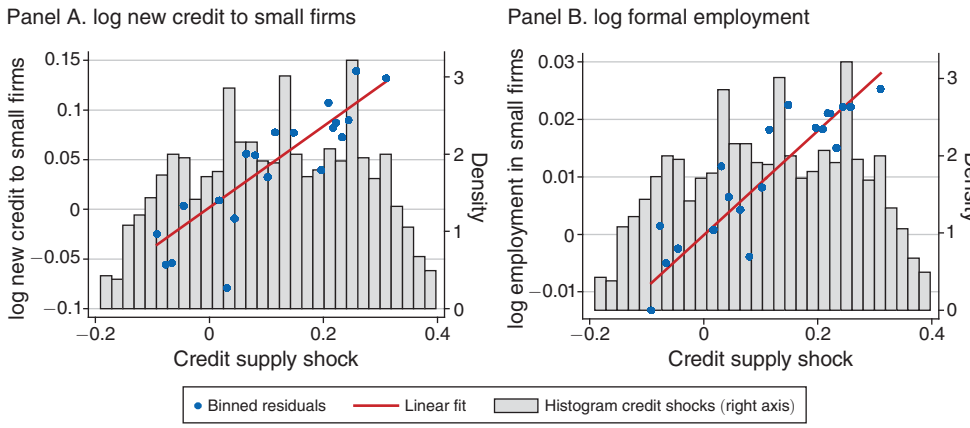

FIGURE 3. MAIN RESULTS

Notes: The figures display binned scatterplots showing the relationship between (panel A) credit supply shocks and new credit to small firms and (panel B) credit supply and log of formal employment in small firms. Data on new credit come from proprietary credit data, and data on formal employment come from the IMSS. The horizontal axis shows the credit supply shocks, which vary at the local labor market–year level. The vertical axis shows (panel A) the residuals of new credit to small firms and (panel B) changes in employment to small firms for each municipality-year cell, controlling for local labor market and year fixed effects and for linear trends interacted with the local labor market controls (fraction of illiterate population, fraction of households with dirt floor, and fraction of households with water). The data are divided into 20 equally sized groups based on the magnitude of the credit shocks. The histogram of credit shocks is shown in gray bars in the backdrop of the figure, and its value is displayed on the right axis. Each point corresponds to the group average of the variable on the vertical axes.

yearly employment in the labor market that receives the shock relative to labor markets with no shocks ($0.16 \times 0.089 \times 100$). This represents 22 percent of the mean yearly employment change in our sample (6.47 percent), suggesting that credit supply shocks can have a considerable impact in LMIEs, where small firms represent a larger share of total employment and are more likely to be credit constrained, as noted above. Panel B in Figure 3 graphically depicts the estimated relationship between credit supply shocks and employment.

C. *Differential Impacts by Shock Magnitude*

Our main specification implicitly assumes that the impact of credit supply shocks on employment is linear. In order to explore the extent to which this assumption is valid in our context and, more generally, how the magnitude of the credit shocks affects employment, we run regressions of the following form:

$$\ln\left(Outcome_{mt}\right) = \sum_k \beta_k \mathbb{I}\left(Z_{m,t} \in (K)\right) + \delta_m + \pi_t + \varepsilon_{m,t}, \tag{5}$$

where $\mathbb{I}\left(Z_{m,t} \in (K)\right)$ are k indicator variables taking a value of one if the credit shock, $Z_{m,t}$, lies in percentile bin K and zero otherwise. Our bins correspond to the following percentiles: 0–15 (credit shock range from -0.595 to -0.066), 15–30 (-0.066 to 0.027), 30–45 (0.027 to 0.089), 45–60 (0.089 to 0.161), 60–75 (0.161 to 0.242), 75–90 (0.242 to 0.308), and 90–100 (0.308 to 0.796). The estimation uses

Table 6—Effect of Credit Supply Shocks on Employment

Dependent variable: log of formal employees in small firms			
	(1)	(2)	(3)
Credit shock$_t$	0.089	0.085	0.095
	(0.031)	(0.039)	(0.038)
Credit shock$_{t-1}$		0.041	0.017
		(0.031)	(0.037)
Credit shock$_{t-2}$			−0.0039
			(0.031)
Observations	2,340	1,872	1,404

Notes: Authors' estimation of equation (4) for the log of formal employees in small firms in each labor market as the dependent variable. Data on formal employment come from the IMSS, and credit shocks are estimated using proprietary credit data. Regressions include local labor markets and year fixed effects. The regression also includes linear trends interacted with the local labor market controls (fraction of illiterate population, fraction of households with dirt floor, and fraction of households with water). The table shows point estimates of the effect of credit supply shocks on changes in employment. Observations are weighted by the number of workers in small firms in each labor market–year cell in 2010. Standard errors are clustered at the local labor market level.

the bin for percentiles 15–30 as the baseline category. All other variables are defined as in equation (4).

Each of the β_k should then be interpreted as the differential change in the outcome for labor markets when the shock lies in bin K relative to the excluded bin. Results of this estimation are presented graphically in Figure 4. Negative shocks are associated with a decrease in formal employment, although this result should be taken with a grain of salt given that we only have one bin with negative values of the shock. Conversely, a positive shock implies an increase in the outcome of interest. This effect appears to increase linearly over the first two positive-valued bins, but the effect then tapers off to a constant impact for percentile bins 45 and higher. This suggests that mild shocks are enough to ease credit constraints and that our results are not driven by outliers.

V. Why Do Credit Shocks Affect Employment in Mexico?

The results presented thus far differ from the findings in Greenstone, Mas, and Nguyen (2020), who employ similar empirical strategies but observe no effects of credit on employment for the United States during times of economic stability. Exploring which features of labor and credit markets could exacerbate the impact of credit on employment may be informative about the generalizability of our findings for other LMIEs when compared to AEs. We focus on institutional characteristics of labor and credit markets that are common in middle-income countries—namely, the relative scarcity of bank credit and the existence of a large informal sector.[16]

[16] Additionally, another explanation for our larger effects for Mexico compared to the United States could be that the magnitude of the shocks is larger in Mexico, such that more variation in the independent variable allows for more precise point estimates, or small credit supply shocks may hinder the stronger effects of larger shocks if

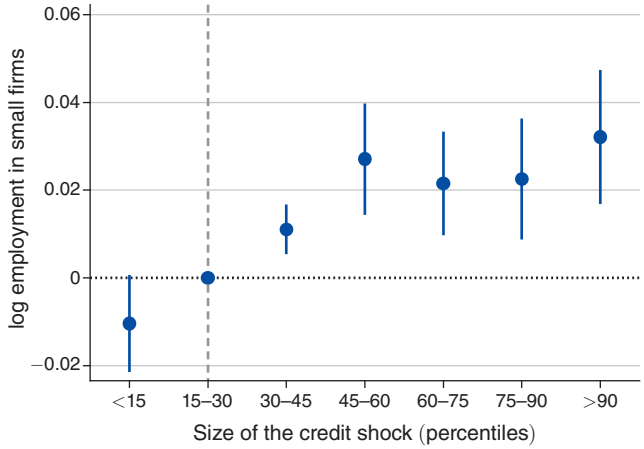

Figure 4. Differential Impacts by Shocks' Magnitude

Notes: Authors' estimation of equation (5) for changes in the log of formal employees in small firms in each labor market as the dependent variable. Data on formal employment come from the IMSS. Regressions include local labor markets and year fixed effects. The regression also includes linear trends interacted with the local labor market controls (fraction of illiterate population, fraction of households with dirt floor, and fraction of households with water). The graph shows point estimates (blue dots) for different magnitudes of the supply size shock, grouped in bins according to the percentile of the shock's distribution: percentiles 0–15 (−0.595, −0.066], 15–30 (−0.066, 0.027], 30–45 (0.027, 0.089], 45–60 (0.089, 0.161], 60–75 (0.161, 0.242], 75–90 (0.242, 0.308], and 90–100 (0.308, 0.796]. The estimation uses as the baseline category percentiles 15–30 because this includes the zero value. Observations are weighted by the number of workers in small firms in each labor market–year cell. The dashed line corresponds to the excluded category. The blue lines represent 90 percent confidence intervals of each of the point estimates.

A. *Credit Supply Shocks in Mexico and the United States*

Financial system differences may explain why credit shocks have larger impacts on the employment of SMEs in Mexico than in the United States during times of economic stability. Credit constraints are more common in Mexico than in a developed country such as the United States, and obtaining good credit conditions, including sufficiently low interest rates, is harder for SMEs in less developed economies. This is in part because the institutional framework and competitive environment make it difficult to overcome asymmetric information in LMIEs, raising switching costs, particularly for SMEs, for credit without collateral and for credit to new firms whose credit history is short (see Section II). In this context, credit-constrained SMEs may benefit comparatively more from a bank-specific idiosyncratic credit supply shock. This, in turn, is consistent with the evidence in Section V, showing

the effects are not linear. To compare the quantitative differences between the shocks in the United States and in Mexico, in online Appendix Table A4 we replicate the US shocks and compare their distribution to that of Mexico, thus better contextualizing the shocks' magnitude. This comparison shows that the mean shock is larger in Mexico than in the United States. This is expected given that the United States analysis covers the financial crisis period, during which the average bank declined new lending by 48 percent (Table 1 in Greenstone, Mas, and Nguyen 2020). Once we demean the shocks, the distribution is very similar to that of Mexico. For example, percentile 10 is −0.19 in our data and −0.23 in the United States, while percentile 90 is 0.196 in our data and 0.22 in the United States. This descriptive table thus indicates that the different results for Mexico and the United States do not seem to be driven by variance in the shocks' magnitude.

that supply-driven credit shocks are good predictors of new credit to SMEs in local labor markets.

In this exercise, we follow the intuition that asymmetric information is more relevant for credit without collateral and for credit to new firms, to those without a recent credit history, and to family firms. We estimate equation (4) using the following dependent variables: the share of credit without collateral and the share of credit to family firms, to young firms, and to firms with no credit lines in the previous year.

Table 7 presents the results. Consistent with our hypotheses, Column 1 shows that the share of credit without collateral increases when a local labor market is exposed to a positive credit supply shock. Specifically, the share of loans without collateral raises by 0.61 percentage points for a 1 percent increase in the credit supply shock. Column 2 shows that credit to family firms increases 0.09 percentage points for a 1 percent increase in the credit supply shock. Column 3 displays a positive and statistically significant impact of the credit supply shocks on the share of credit to young firms, equal to 0.15 percentage points for a 1 percent increase in the credit supply shock. Column 4 shows that the share of credit going to firms with no bank relationship in the previous year also increases significantly by 0.11 percentages points. Importantly, this last result indicates that part of the effect of credit supply shocks on employment is originated by firms that did not have access to the financial system.

Finally, since the characteristics of banking systems in LMIEs result in both credit rationing and higher interest rates for small firms, we further extend the analysis to investigate the impact of credit supply shocks on interest rates. Column 5 shows that these shocks have no impact on the interest rate that banks in Mexico charge to small firms. This is likely due to an offsetting effect: although an increase in credit supply should decrease the interest rate for some firms, this is compensated by an increase in credit availability to riskier firms that usually face higher interest rates.

Given these results, it could be that firms that experience a larger credit shock are also relatively smaller and thus arguably less productive. The credit shocks may thus have different effects for high- and low-skilled employment. Recent literature documents that changes in credit supply in AEs might not affect high- and low-skill workers' employment in the same way. For example, Moser et al. (2022) exploit a negative credit supply shock in Germany and find that the shock mostly affected low-wage employment due to firms financing their labor expenses using debt and adjusting to the shock through reducing low-skilled employment and adjusting high-skilled wages. In light of these findings, in this subsection we explore whether credit shocks have heterogeneous impacts on employment across workers earning more than two minimum wages and those earning less.[17] While this distinction does not perfectly match the idea of high- and low-skill workers in the United States, it does provide a suitable categorization of labor markets in Mexico.

Table 8 presents the results of our main specification using the log of low-wage formal employment (column 1) and of high-wage formal employment (column 2) in each local labor market as the dependent variables. We find that the results on employment are concentrated in low-wage employment, consistent with the larger

[17] The data on formal employment do not include information on workers' educational level, preventing us from performing the analysis by skill level.

Table 7—Effect on Credit Characteristics

Dependent variable:	Share of credit without collateral	Share of credit to family firms	Share of credit to new nonfamily firms	Share of credit to firms with no credit in the previous year	Interest rate
	(1)	(2)	(3)	(4)	(5)
Credit shock$_t$	0.611	0.090	0.149	0.113	0.014
	(0.117)	(0.045)	(0.064)	(0.054)	(0.058)
Observations	2,340	2,095	2,093	2,340	2,335

Notes: Authors' estimation of equation (4) for main outcome variables. Regressions include local labor markets and year fixed effects. Data on new credit characteristics come from proprietary credit data. The regression also includes linear trends interacted with the local labor market controls (fraction of illiterate population, fraction of households with dirt floor, and fraction of households with water). The first column shows results for the share of total new credit without collateral. The second column shows results for the share of new credit directed to family firms. The third column further differentiates nonfamily firms into new firms (5 years old or less) and older firms (more than 5 years old), the only type of firm for which the date of creation is available in the data (not available for family firms). Column 4 presents the effect of the share of new credit to firms that did not have any outstanding credit the year before. The last column contains results for the log of the average interest rate. Columns 2 and 3 restrict the sample to local labor market with presence of nonfamily firms. The table shows point estimates of the effect of credit supply shocks on changes in credit characteristics. Observations are weighted by the number of workers in small firms in each labor market–year cell in 2010. Standard errors are clustered at the local labor market level.

effect on new credit being concentrated in firms that are more likely to be credit constrained. These firms are presumably more intensive in low-skilled labor.[18] Online Appendix Table A6 further shows a negative effect on average wages. This is consistent with most of the changes in employment being concentrated in low-wage employment. In fact, average wages even decline for low-wage employment, indicating that new hires must be very close to the minimum wage.[19]

Credit constraints may also be larger in regions with relatively low access to credit but with enough mass of formal firms to take advantage of the shock. To explore this, online Appendix Table A7 shows the main results interacting the credit supply shock with indicator variables for the degree of credit availability in a region in the baseline year, as measured by credit per worker. Specifically, we include interactions with dummy variables indicating the intensity of access to credit in 2010, computing percentiles 33 and 66, such that high credit means markets with more than $17,694 (real Mexican pesos) per formal worker, medium credit between $6,426 and $17,694, and low credit less than $6,426. The base category is high credit per worker. Consistent with this hypothesis, we find that the largest effect is concentrated in regions with medium levels of credit availability. In regions with more credit availability, the effects are still significantly different from zero, though in such regions it may be

[18] In Mexico, the size of the firm is negatively correlated with the share of low-wage employees. In 2010, around 70 percent of workers employed in firms with fewer than 10 employees earned less than twice the minimum wage. This share declines to 50 percent for firms between 10 and 50 employees, 30 percent for those between 50 and 250 employees, and 20 percent for more than 250 employees. In our sample, family firms have fewer average employees (10 employees in 2010) than nonfamily firms (43) (see online Appendix Table A2). In addition, young nonfamily firms (30) and firms without a credit history (27) tend to be smaller than older nonfamily firms (44) and firms with a recent credit history (40).

[19] Online Appendix Table A5 also shows the employment effects for each gender separately. We find similar effects for men and women, with point estimates that are slightly larger for the former, though they are not statistically significantly different.

Table 8—Effect of Credit Supply Shocks on Low- and High-Wage Employment

Dependent variable: log of formal employees in small firms		
	Low-wage employees	High-wage employees
	(1)	(2)
Credit shock$_t$	0.157	0.016
	(0.038)	(0.038)
Observations	2,340	2,333

Notes: Authors' estimation of equation (4) for log of formal employees earning less and more than two minimum wages in small firms in each labor market as the dependent variable. Data on formal employment come from the IMSS, and credit shocks are estimated using proprietary credit data. Regressions include local labor markets and year fixed effects. The regression also includes linear trends interacted with the local labor market controls (fraction of illiterate population, fraction of households with dirt floor, and fraction of households with water). The table shows point estimates of the effect of credit supply shocks on changes in low-wage employment and high-wage employment in columns 1 and 2, respectively. Observations are weighted by the number of workers in small firms in each labor market–year cell in 2010. Standard errors are clustered at the local labor market level.

easier for firms to switch banks and to overcome bank-specific credit supply changes. In areas with low credit availability, there might not be enough firms at the margin to take advantage of the credit supply shock, such that the impact on new credit granted to firms and, thus, on employment is lower. These results provide additional evidence that credit supply shocks on employment may be larger for Mexico than the United States because firms tend to be more credit constrained.

Taken together, these results suggest that credit supply shocks in Mexico have an effect on employment because they help firms ease credit constraints, and that this is particularly relevant for the firms that banks consider to be riskier. These firms are more likely to grow when exposed to a positive credit supply shock by hiring low-skilled workers and paying them close to the minimum wage.

B. *The Importance of the Informal Sector*

One of the most salient differences between the labor markets of LMIEs and AEs concerns the size of their informal sector. In this section, we explore whether the increase in formal employment associated with credit supply shocks can be explained by the existence of a considerable mass of potential formal employees working in informal jobs. In this view, the same positive credit supply shock in a LMIE, shifting labor demand upward, would have larger impacts on employment levels since the labor supply in the formal sector is more elastic.[20]

To explore the role of labor informality, we exploit the ENOE and investigate whether the increase in formal employment is driven by a decrease in informal employment. Table 9 presents the results of our main specification, using the measures of formal and informal employment in SMEs obtained from the ENOE as

[20] In the classical development literature on dual labor markets, it is often assumed that people are working informally because they are unable to work formally. Thus, supply in the formal sector is perfectly elastic, and formal employment is solely determined by labor demand. More recent papers consider that many informal jobs might be preferred to other low-paid formal jobs, such that labor supply in the formal sector is not necessarily more elastic in countries with a large informal sector. See Fields (2011) for further details.

Table 9—Effect of Credit Supply Shocks (ENOE Sample)

Credit and formal employment restricted to localities for which the ENOE sample is representative					
Dependent variable:	log new credit (1)	log employment (IMSS) (2)	log total employment (ENOE) (3)	log informal employment (ENOE) (4)	log formal employment (ENOE) (5)
Credit shock$_t$	0.750	0.093	−0.018	−0.043	0.369
	(0.341)	(0.081)	(0.145)	(0.160)	(0.298)
Observations	175	175	175	175	175

Notes: Authors' estimation of equation (4) for different outcome variables. Data on new credit come from proprietary credit data, and data on employment come from the IMSS or ENOE. Regressions include local labor markets and year fixed effects. The regression also includes linear trends interacted with the local labor market controls (fraction of illiterate population, fraction of households with dirt floor, and fraction of households with water). The table shows point estimates of the effect of credit supply shocks on changes in credit and employment. Standard errors are clustered at the local labor market level.

dependent variables, and compares them with the point estimates from the IMSS sample after restricting to localities where the ENOE is representative (grouped in local labor markets). Unfortunately, this survey is only representative of a small number of large cities, losing more than 90 percent of the observations relative to the estimation with the IMSS database.[21] Although the loss of observations in this dataset does not give us enough statistical precision, the point estimate in the IMSS data in column 2 remains of similar magnitude to that in Table 6, though it is no longer statistically significant. Our estimates are even noisier for the ENOE sample, but we do find a point estimate close to zero for total employment (column 3), a negative point estimate for informal employment (column 4), and a positive point estimate for formal employment (column 5), none of them statistically significantly different from zero. These findings suggest that the existence of a large informal sector in Mexico could in part explain the strong effects of credit supply on formal employment, given that the results in this table are consistent with workers moving from the informal to the formal sector. That said, caution should be taken in interpreting these results, as the imprecision of our estimates means that we cannot reject that increases in credit supply do not reduce informal employment.

To shed further light on whether a large informal sector underlies the impact of credit on formal employment, we assess whether the effects of credit supply shocks vary by a labor market's baseline share of informal workers using the IMSS data (Table 10). Specifically, the regressions now include interactions with binary variables indicating the informality rate by terciles in a local labor market in 2010, according to the economic census. In contrast with ENOE, this information from the census is available only for 2010 but is representative of all labor markets included in our sample. A low informality rate indicates markets with a greater than 30 percent share of formal workers; medium informality, between 12 and 30

[21] Because the ENOE sample is concentrated in large urban localities, coverage is larger in terms of share of workers than with respect to the number of observations: it covers 40 percent of formal workers in 2010.

Table 10—Heterogeneity By Informality Level

Dependent variable:	log new credit	log formal employees	log low-wage employees	log high-wage employees
	(1)	(2)	(3)	(4)
Informality (*Base category: low informality*)				
Credit shock$_t$	0.314	0.089	0.152	0.019
	(0.190)	(0.032)	(0.038)	(0.039)
Credit shock$_t$ × MediumI$_m$	0.348	−0.001	−0.013	0.018
	(0.100)	(0.023)	(0.027)	(0.027)
Credit shock$_t$ × HighI$_m$	−0.097	−0.057	−0.049	−0.070
	(0.159)	(0.024)	(0.031)	(0.035)

Notes: Authors' estimation of equation (4) for main outcome variables. Data on new credit come from proprietary credit data, and data on formal employment come from the IMSS. Regressions include local labor markets and year fixed effects. The regression also includes linear trends interacted with the local labor market controls (fraction of illiterate population, fraction of households with dirt floor, and fraction of households with water). The regressions include interactions with dummies indicating the level of informality rate in a local labor market in 2010 according to the economic census: MediumI$_m$ (HighI$_m$) is an indicative variable that takes the value of 1 if labor market m has medium (high) informality in the baseline year and zero otherwise. The dummy variables are constructed using percentiles 33 and 66, such that low informality indicates markets with a share of formal workers that is greater than 30 percent, medium informality between 12 and 30 percent, and high informality less than 12 percent. The table shows point estimates of the effect of credit supply shocks on changes in credit and employment. Observations are weighted by the number of workers in small firms in each labor market–year cell in 2010. Standard errors are clustered at the local labor market level.

percent; and high informality, a less than 12 percent share of formal workers. It is well known that a large share of the labor force in Mexico works in the informal sector—around 60 percent in the 2010 census. We observe that the impacts of the credit shocks on employment are present across all labor markets but are larger for markets with low and medium informality rates.[22] This evidence suggests that while our results might partly be driven by informal workers moving to formal jobs, this is not the only mechanism: in regions where informality is more pronounced, we also find smaller changes in formal employment.

VI. Robustness Checks

A. *Dynamic Effects*

Tables 5 and 6 showed that there are no dynamic effects of credit shocks in previous years on new credit and employment. However, there still could be dynamic effects within a year, such that employment at the end of the year depends on credit supply shocks in different months or quarters. Although interesting, the data at hand prevent us from assessing these within-year dynamics in detail. To precisely estimate the bank-specific component of the credit supply shock (equation (1)), it needs to be

[22] The total employment effect in regions with high informality rates is no longer statistically significantly different from zero. This result is consistent with low labor enforcement in these regions, where credit supply shocks might increase total employment but have less effect on formal employment because firms are less likely to comply with labor market regulations (Samaniego and Fernández Bujanda 2020). If this is the case, our results might show a lower bound of the true effect on total employment in these regions.

distinguished from the local labor market demand. Doing so requires having enough data on log changes in new credit granted by each bank in different local labor markets during the reference period. The shorter the considered period, the noisier these log changes become, and more bank–local labor market cells with zero values are dropped from the estimation. Therefore, the bank-specific components used to estimate credit supply shocks with monthly and even quarterly data are no longer sufficiently informative (i.e., the shocks are not related with observed changes in new credit). Because there is a trade-off between working with higher-frequency data and the precision of our estimates, all of our main results use annual data. The only exception is online Appendix Table A8, which shows our main results using biannual data incorporating three lags. We observe that there are some dynamic effects of credit supply shocks on new credit. In the case of employment, the only estimates that remain significant in all of our regressions are those of the contemporaneous shocks. The point estimates of the lagged credit supply shock are positive but noisily estimated. The sum of the coefficients in the case of employment is lower than our estimation using annual data, indicating that the results in online Appendix Table A8 could be biased downward due to measurement error in our treatment variable. We interpret these results as suggestive evidence that there could be dynamic effects within a given year, but our data do not allow to estimate credit supply shocks with enough precision at the biannual or quarterly level to identify them.

B. *Differences in the Unit of Analysis*

A potential driver of the effects found in this paper may be related to having chosen local labor markets as our relevant unit of analysis. This contrasts with Greenstone, Mas, and Nguyen (2020), who use counties as the unit of observation instead of, for example, commuting zones. In order to explore whether this is the case, we replicate our main results, recalculating all explanatory and dependent variables at the municipality level, the finest administrative partition that our data allow. Table 11 presents our findings, where panel A reproduces the estimates from previous sections (at the labor market level), and panel B displays the analogous estimates when instead using municipalities as the unit of analysis. For the two main outcomes of interest, total new credit to small firms (column 1) and formal employment in small firms (column 2), the coefficients associated with the credit shocks are positive and similar in magnitude (we cannot reject that they are equal). Nonetheless, the estimates at the municipality level are noisier, despite the increase in the number of observations, and smaller in absolute value. The only exception is the coefficient for high-wage employment, which we interpret with caution because it is only significant at the 10 percent level.[23]

These findings suggest that municipalities participating in the same labor market may also be affected by a credit supply shock in neighboring municipalities

[23] It is possible that labor supply for high-wage employment is less elastic than low-wage employment at the labor market level. Thus, benefited municipalities may increase this type of employment by absorbing high-wage employment from neighboring municipalities participating in the same labor market (Barrot and Nanda 2020). If this is the case, the SUTVA assumption is violated because nontreated units are also negatively affected by the shock (see Section IIE), so that the point estimate for high-wage employment is upward biased.

Table 11—Effect of Credit Supply Shocks in Local Labor Markets and Municipalities

	log new credit	log formal employees	log low-wage employees	log high-wage employees
	(1)	(2)	(3)	(4)
Panel A. Local labor markets as the unit of analysis				
Credit shock$_t$	0.413	0.089	0.157	0.016
	(0.187)	(0.031)	(0.038)	(0.038)
Observations	2,340	2,340	2,340	2,333
Panel B. Municipalities as the unit of analysis				
Credit shock$_t$	0.421	0.055	0.059	0.050
	(0.176)	(0.015)	(0.030)	(0.026)
Observations	4,590	4,587	4,585	4,587

Notes: Authors' estimation of equation (4) for different outcome variables using local labor markets (panel A) or municipalities (panel B) as the unit of analysis. Data on new credit come from proprietary credit data, and data on formal employment come from the IMSS. Regressions include region and year fixed effects. The regression also includes linear trends interacted with the local labor market controls (fraction of illiterate population, fraction of households with dirt floor, and fraction of households with water). The table shows point estimates of the effect of credit supply shocks on changes in credit and employment. Observations are weighted by the number of workers in small firms in each labor market–year cell in 2010. Standard errors are clustered at the local labor market level.

within the same labor market and are therefore an inadequate (contaminated) control group. This is consistent with Huber (2018), who finds that the indirect effects of credit supply shocks can be large, and depend on the intensity of local agglomeration spillovers (e.g., knowledge spillovers, transport costs, the quality of workers in the local labor market) and the effect on local aggregate demand, which may transgress municipalities' administrative boundaries.

Broadly, this assessment shows that choosing labor markets as our relevant unit of observation is an important methodological decision. It results in more precise estimates that are larger in magnitude and helps explain part of the discrepancy between our results for Mexico and those for the United States. That said, this choice is certainly not the only reason why we identify effects of credit supply shocks on employment in the Mexican context during times of economic stability.

C. *Alternative Specifications*

Although the credit supply shocks estimated by equation (3) are arguably exogenous from other factors that could alter local labor supply and local labor demand, we specifically test it in this section. First, we add time trends interacted with demographic variables to our specification to allow for differential trends in each labor market depending on the characteristics of its population (education, age, access to basic services, etc.). Second, we test whether our results are robust to adding local labor demand shocks as a control. Specifically, the latter measures the predicted change in employment in each labor market given its industry composition and the national employment trends in each industry, and could be related to the baseline bank shares in each labor market. Finally, we explore whether our results change

when controlling for homicide rates as a proxy for local violence, which could be a relevant economic variable in the Mexican context affecting both employment and credit at the labor market level.

Table 12 shows the results from these exercises, where panel A replicates the main results without interacting any time trends with local characteristics, panel B includes time trends interacted with local characteristics, panel C includes labor demand shocks as controls, and panel D controls for homicide rates. While the inclusion of municipality demographic characteristics interacted with time trends (panel B) allows us to gain precision, the point estimates are not statistically different from not incorporating any controls (panel A) and are similar to those of our main estimates.[24] We interpret this as evidence that the estimated credit supply shocks do not follow any particular trend across municipalities associated with other underlying characteristics of these municipalities that could bias our estimates.

In panel C, the local demand shocks are constructed by interacting baseline differences in industrial composition with national percentage changes in industry employment (excluding the labor market of interest).[25] That is,

$$LDS_{mt} = \sum_{i=1}^{I} \theta_{m,i,t-1} \left(\frac{\pi_{-m,i,t} - \pi_{-m,i,t-1}}{\pi_{-m,i,t-1}} \right), \tag{6}$$

where LDS_{mt} is the predicted local demand shock in local labor market m for year t, $\theta_{m,i,t-1}$ are industry i employment shares in baseline year $t-1$, and $\pi_{-m,i,t}$ is the number of workers at the national level in industry i in year t, excluding local labor market m. This index can be interpreted as the predicted yearly changes in employment due to the local industry composition of local labor market m and the industry-specific national employment trends. If the estimated credit supply shocks were related to the industry composition of local labor markets due to, for example, banks and labor markets specializing in some industries, our estimates might be picking up these baseline differences, biasing the results. In other words, the bank shares $s_{b,m,t}$ used to construct the credit shock $Z_{m,t}$ could be related to the industry composition in each labor market.

The endogeneity of the bank shares can, however, be problematic if identification is based on a before-and-after comparison with, for example, a cross-section regression comparing changes in treated and untreated units with different baseline shares (Goldsmith-Pinkham, Sorkin, and Swift 2020). In our case, with multiple time periods, the exogeneity of the shocks can provide identification even when shares are endogenous (Borusyak, Hull, and Jaravel 2022). Intuitively, because of the panel structure of our data, we can control for any time-invariant local labor market differences (fixed effects), and the variation in the estimated credit supply shocks over time is generated by the shocks and not by the shares.[26] Thus, we do not

[24] The demographics characteristics used in the regression are displayed in Table 3 and include the share of the population in different age intervals (0–14 and 15–64), share of migrants, share of indigenous population, share with different schooling (illiterate, primary and secondary), share not covered with social insurance, and share living in households with electricity.

[25] This Bartik-style instrument has been widely used in the literature. For a recent paper, see Notowidigdo (2020).

[26] Borusyak, Hull, and Jaravel (2022) show that unit fixed effects play a key role in shift-share estimates by isolating variation in shocks over time.

Table 12—Robustness Checks

Dependent variable:	log new credit	log formal employees	log low-wage employees	log high-wage employees
	(1)	(2)	(3)	(4)
Panel A. No controls				
Credit shock$_t$	0.396	0.101	0.176	0.019
	(0.183)	(0.033)	(0.041)	(0.038)
Panel B. Controlling for lineal trends using all municipality characteristics				
Credit shock$_t$	0.472	0.062	0.115	0.006
	(0.173)	(0.028)	(0.033)	(0.037)
Panel C. Controlling for local labor demand shocks				
Credit shock$_t$	0.415	0.090	0.157	0.018
	(0.188)	(0.031)	(0.038)	(0.038)
Panel D. Controlling for homicide rate				
Credit shock$_t$	0.407	0.086	0.154	0.013
	(0.187)	(0.031)	(0.039)	(0.038)

Notes: Authors' estimation of equation (4) for main outcome variables. Data on new credit come from proprietary credit data; data on formal employment come from the IMSS; municipality characteristics are obtained from 2010 census data; homicide rates are calculated using homicide data from the Secretariado Ejecutivo del Sistema Nacional de Seguridad Pública (2020). Regressions include local labor markets and year fixed effects. Panel A does not control for any municipality characteristics. Panel B controls for all the local labor market demographic characteristics displayed in Table 2 interacted with year dummies. These characteristics include the share of the population in different age intervals (0–14 and 15–64), share of migrants, share of indigenous population, share of different schooling levels (illiterate, primary, and secondary), share not covered by social insurance, and share living in households with electricity. Panel C controls for local labor demand shocks estimated by equation (3), which controls for the predicted change in employment given the industry composition of each local labor market and the national employment trends in each industry. Panel D controls for the homicide rate per 10,000 people in each local labor market–year cell. The table shows point estimates of the effect of credit supply shocks on changes in credit and employment. Observations are weighted by the number of workers in small firms in each labor market–year cell in 2010. Standard errors are clustered at the local labor market level.

expect our estimates to change with the introduction of local labor demand shocks. As expected, panel C shows that our results are robust to the inclusion of local labor demand shocks: the magnitude of our point estimates and its statistical significance is practically the same as that observed in Tables 5 and 6.

Finally, panel D controls for the homicide rate in each local labor market.[27] The rationale for this exercise is that increases in local violence could be an omitted variable that may decrease both firm credit and formal employment, such that not controlling for this factor could bias our results.[28] However, as we do not regress employment on actual credit but rather use credit supply shocks, these shocks should be uncorrelated with local labor market conditions. That is, the correlation of credit supply shocks and the omitted variable (homicide rate) should be approximately

[27] Homicide data come from the Secretariado Ejecutivo del Sistema Nacional de Seguridad Pública (2020), which is presented at the municipality level, and we aggregate the data at the labor market–year level.

[28] Utar (2021) finds that the increases in violence in Mexico, arising from the drug war between 2005 and 2010, reduced local demand and labor supply. Arguably, banks might consider it riskier to lend in markets with increases in violence. This is consistent with Bank of Mexico (2020b), which finds that increases in the local homicide rate are correlated with a raise in the interest rate charged to firms operating in that region.

zero. The results in panel D confirm this: our main estimates remain the same after controlling for the homicide rate. We interpret these findings as additional suggestive evidence that the credit shocks we estimate are completely exogenous to the local labor market conditions.

D. *Credit and Employment in Large Firms as a Placebo*

As mentioned in Section I, only 3 percent of large firms report being credit constrained in Mexico, as opposed to 37 percent of SME firms. Because large firms are less credit constrained and can (arguably) switch across banks more easily, they should be much less affected by these shocks. We formally test whether the evidence is consistent with this conjecture in Table 13. Panel A presents results relative to the effect of the estimated credit supply shocks on large-firm employment. The credit supply shocks we use in panel A only consider, however, data on credit to small firms and may therefore not be representative of changes in credit supply to large firms if the bank relationships of the latter tend to differ from those of small firms. Panel B accordingly presents results estimating an alternative measure of exposure to credit supply shocks that is estimated using only new credit directed to large firms.

As expected, in both panels, the credit supply shocks have no statistically significant effects on large firms' employment. Note that the sample size for large firms is smaller than that for small firms in Table 6, given that they are present in a smaller number of local labor markets.[29] However, if we restrict the analysis in Table 6 to small firms in local labor markets with large-firm presence, we obtain a point estimate for formal employment equal to 0.127 and a standard deviation of 0.048. This result indicates that even in this restricted sample, there is still an effect of credit supply shocks on small firms' formal employment, but not one on that of large firms.

Online Appendix Table A9 shows additional results for large firms, redefining the outcomes to include all observations. Panel A shows results when outcomes are considered as shares—that is, large firms' shares of new credit and employment in each labor market. Although the outcome variable has several zero values (52 percent of all observations), results across the table show no significant effect of credit supply shocks on these shares when all observations are considered. The lack of statistically significant results in panel A could be driven by an increase in outcomes for both large firms and SMEs. To explore if this is the case, panel B considers only the outcomes of large firms normalized by the regional economically active population (aged 15–64). For comparison, panel C repeats this normalization only for SMEs. Consistent with Table 13, results in panel B show no significant effect across the columns for large firms. On the contrary, results for small firms are robust to normalizing the dependent variable in per capita values, as opposed to using the log transformation. These additional exercises confirm the lack of effect of credit supply shocks for large firms regardless of how we deal with zero values in the dependent variable for these firms.

[29] As in the case of small firms, we restrict the sample to the labor markets with nonzero observations for new credit lines to large firms and for the number of employees in large firms.

Table 13—Results for Large Firms

Dependent variable:	log new credit	log formal employees	log low-wage employees	log high-wage employees
	(1)	(2)	(3)	(4)
Panel A. Using shocks to small firms				
Credit shock$_t$	1.176	0.187	0.351	0.130
	(1.656)	(0.122)	(0.243)	(0.138)
Observations	508	508	505	508
Panel B. Using shocks to large firms				
Credit shock$_t$	0.306	0.026	0.005	0.023
	(0.272)	(0.029)	(0.082)	(0.027)
Observations	508	508	505	508

Notes: Authors' estimation of equation (4) for main outcome variables. Regressions include local labor markets and year fixed effects. Data on new credit come from proprietary credit data, and data on formal employment come from the IMSS. The regression also includes linear trends interacted with the local labor market controls (fraction of illiterate population, fraction of households with dirt floor, and fraction of households with water). The table shows point estimates of the effect of credit supply shocks on changes in employment for large firms (more than 250 employees). Observations are weighted by the number of workers in large firms in each labor market–year cell in 2010. Panel A takes as independent variable the credit supply shocks estimated for small and medium firms and estimates its effects on large-firm employment. Panel B considers credit shocks specifically estimated for large firms. The sample is restricted to the 116 local labor markets for which we have information on credit to large firms. Standard errors are clustered at the local labor market level.

VII. Conclusions

Credit supply is commonly viewed as a key determinant of employment creation in LMIEs, yet identification of this causal link has proved challenging due to data availability restrictions. In this paper, we measure the causal relationship between credit supply shocks to SMEs and employment in Mexico. We exploit an extraordinarily detailed dataset containing loan-level information on the universe of loans extended by commercial banks to private firms in Mexico for the 2010–2015 period, coupled with publicly available data on the number of employees reporting formal wages to the IMSS. To construct our explanatory variable, we first estimate idiosyncratic changes in lending from each bank that cannot be explained by changes in demand for credit across labor markets. We then calculate a labor-market level weighted average of these credit shocks, in which we use the market share of each bank in each labor market as weights.

We document that credit supply shocks have a large and significant impact on formal employment in Mexico. This finding contrasts with recent work focusing on the relationship between credit and employment in developed economies, suggesting that credit supply shocks have a greater effect in LMIEs where SMEs represent a larger share of total employment and are more credit constrained. A positive credit shock equal to 1 standard deviation increases yearly employment changes by 1.42 percentage points, which represents 22 percent of the mean yearly employment change in our sample (6.47 percent). Moreover, the credit supply shocks we estimate are associated with increases in loans without collateral, loans to new firms, to

family firms, and to firms without recent credit history; that is, the type of enterprises that are more strongly subject to informational asymmetries and credit constraints.

In sum, our findings indicate that these asymmetries and constraints are particularly pertinent for SMEs in LMIEs. The results are furthermore relevant to understanding the economic effects of COVID-19 and recovery in the post-pandemic period. While firms of all sizes have been negatively affected by COVID-19, sales shrank more and liquidity drained faster for SMEs than for large firms (Adian et al. 2020). At the same time, LMIEs have limited fiscal space to support SMEs, though it is precisely in these countries that SMEs play a more prominent role in job creation. The results of this paper suggest that, given the current context, policy measures that relax credit constrains and facilitate good credit conditions for SMEs could play a central role in aiding the recovery of formal employment in LMIEs.

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